

Venkata Raja Pavan Kumar Reddy Valila

Java Backend Developer | Spring Boot | REST APIs | SQL | AWS | AI-Integrated Backend Systems
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SUMMARY

Java Backend Engineer with 2+ years of experience building secure, scalable backend systems using Java, Spring Boot, and Spring Security, currently delivering production support and modernization work at Tata Consultancy Services. Strong foundation in Data Structures & Algorithms, REST API design, and SQL-based data modeling. Hands-on experience integrating AI capabilities into backend services using Spring AI — including prompt design and structuring LLM inputs/outputs for real application features — reinforced by the Claude Certified Architect (Foundations) credential. Actively building toward product-based backend and AI-augmented engineering roles.

TECHNICAL SKILLS

- **Programming:** Java (Primary), SQL
- **Core Java:** OOPs, Collections, Multithreading, Exception Handling, Java 8 (Streams, Lambdas, Optional)
- **Backend Technologies:** Spring Boot, Spring Security, REST APIs, JWT, JPA, Hibernate
- **AI-Integrated Development:** Spring AI, Prompt Engineering, LLM Input/Output Design, Generative AI Architecture Fundamentals
- **Architecture:** Microservices, Layered Architecture, MVC, API Design
- **Data Structures & Algorithms:** Problem Solving, Complexity Analysis
- **Databases:** MySQL, H2
- **Tools:** Git, GitHub, IntelliJ IDEA, Postman

CERTIFICATIONS

- **Claude Certified Architect - Foundations** — Anthropic | Issued Aug 2026 · Expires Aug 2027
- **Microsoft Certified: Azure Security Engineer Associate** — Microsoft | Issued Oct 2024 · Expires Oct 2026
- **Oracle Certified Professional, Java SE 8 Programmer I (OCP)** — Oracle | Issued Nov 2024

PROJECTS

AI-Powered Productivity Backend System | *Java, Spring Boot, Spring Security, JWT, SQL, REST APIs, Spring AI*

- Designed and developed a scalable Spring Boot backend exposing 15+ REST APIs for task management, goals, and productivity tracking.
- Implemented JWT-based authentication and role-based access control (RBAC) to secure application data.
- Applied Core Java (Streams, Collections) and DSA concepts to optimize task prioritization and analytics.
- Integrated Spring AI to generate personalized productivity insights — designed the prompt structure for translating user activity data into model input and parsed the LLM's output into actionable, ranked recommendations.
- Ensured scalability using modular architecture, centralized exception handling, and API design best practices.

Secure E-Commerce Backend System | *Java, Spring Boot, Spring Security, JPA, SQL, REST APIs*

- Developed backend services for users, products, orders, and inventory using Spring Boot.
- Secured 10+ REST APIs using Spring Security with JWT-based authentication and RBAC.
- Optimized API performance using Java Collections and DSA techniques, improving response time by 20%.
- Designed relational database schemas using Spring Data JPA and SQL with proper entity relationships.
- Improved system reliability with validation, logging, and centralized exception handling.

EXPERIENCE

Assistant System Engineer — Tata Consultancy Services (TCS)

Apr 2024 – Present

India

- Contributed to legacy system modernization and cloud migration, improving performance by 10%.

- Own mission-critical end-to-end testing for 5+ enterprise applications, ensuring reliable backend data processing and delivery; recognized as a Best Performer for achieving 100% on-time delivery across stakeholder-driven deadlines.
- Resolved 20+ production issues, improving system stability and data flow consistency.
- Collaborated with cross-functional teams to understand backend workflows, APIs, and system integrations.
- Built hands-on experience in Java, Spring Boot, REST APIs, SQL, and Spring Security (JWT-based authentication).

Backend Developer — Freelance Project (Academic)

Jan 2023 – Mar 2023

Remote

- Developed backend services using PHP and SQL, reducing manual processing by 25%.
- Implemented RESTful APIs and role-based access control for secure data handling.
- Optimized performance by resolving 10+ backend issues through debugging and testing.

EDUCATION

JNTUA College of Engineering

Mar 2023

Bachelor of Technology in Electrical and Electronics Engineering (CGPA: 8.8 / 10.0)

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Microprocessors & Microcontrollers, Electrical Circuits, Power Systems